



SERENDIB COLLEGE,
GAMPOLA
FIRST TERM TEST - 2026
Grade 11
Mathematics 1

Time :3 Hours

Part A

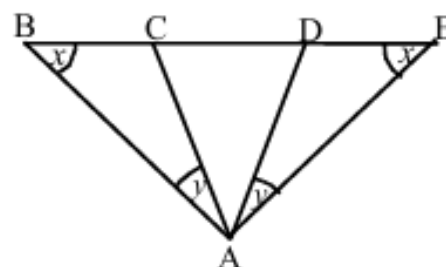
1. Find the least common multiple of the two algebraic expressions $4a^2b$, $3ab$

2. Value of $\sqrt{76}$ lies between in which two integers.

3. Simplify $\frac{1}{2x} - \frac{1}{2x-1}$

4. If $\lg x = 3$, find the value of x .

5. Name 2 isosceles triangles based on the information given in the diagram.

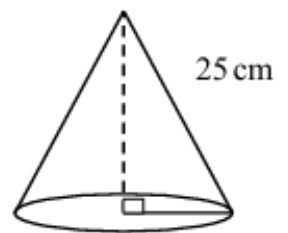


6. Factorize $2x^2 - 50$

7. Solve $x^2 - 3x - 18 = 0$

8. 15% of VAT was charged on an electricity bill of Rs. 3600. Calculate the amount to be paid.

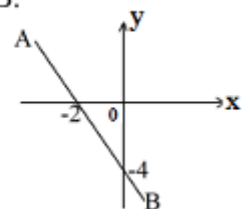
9. The curved surface area of the cone given in the figure is 1100 cm^2 . Find the radius of the Base. (Radius of the cone is r slanted height is l . Curved surface area of the cone is πrl . $\pi = \frac{22}{7}$)



10. Simplify $\frac{12xy^2}{5b} \div \frac{18xy}{25b}$

11. In a box there are 6 card numbered from 1 to 6. What is the probability of a card taken randomly from the box written with a prime number?

12. According to the data given in the diagram, write the equation of the straight line AB.

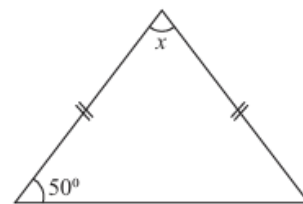


13. It takes 4 men 9 days to complete a certain task. How many men are needed to complete the same task within 6 days?

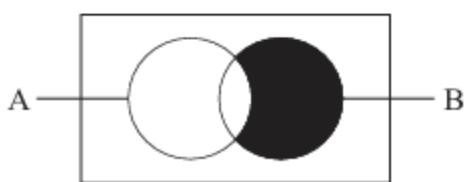
14. Solve

$$\frac{3+x}{2} = 5$$

15. According to the information in the figure, Find the value of x .

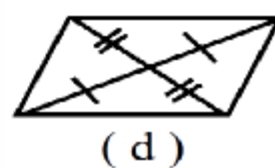
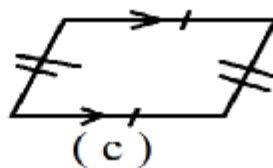
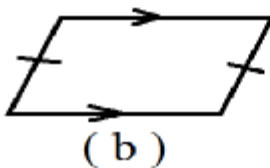
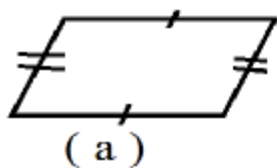


16. Write the shaded region in the venn diagram in set notation.

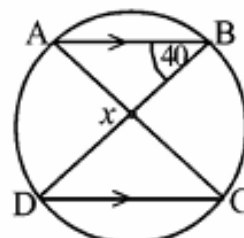


17. A vehicle travels at a speed of 72 km h^{-1} . Find the distance travelled in 25 minutes.

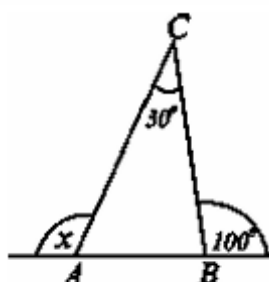
18. According to the data given in the figure, select and underline diagram which is not a parallelogram.



19. Two chords of the circles AB and CD are intersected at X. Find the magnitude of $\angle DXC$, using the given information.



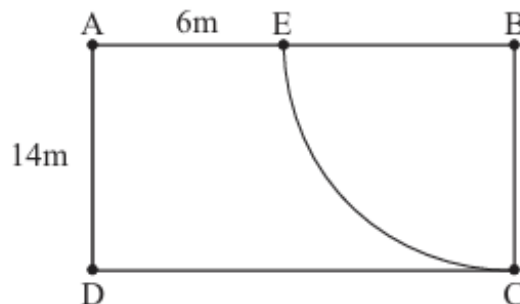
20. By using the given information find the value of x



Part B

1. A certain tea factory exported $\frac{5}{8}$ of what was produced in a certain month and $\frac{1}{6}$ was issued to the local wholesale market.
- (i) What fraction of the total production that exported and issued to local wholesale market ?
- (ii) $\frac{1}{4}$ of the remaining amount was distributed among factory workers. What fraction of the total stock that distributed ?
- (iii) Finally the remaining quantity was distributed to the shop in the factory premises. If that given quantity was 600 kg, find the amount of tea exported.
- (iv) If the 1 kg of tea sold at Rs. 850 in the local market. Find the total amount of money from the sale.

2. In the rectangular shaped land, BCE is a pond. Grass is grown in the remaining portion AECD.



- i) Find the perimeter of the land.
- ii) Find the perimeter of the land where grass is grown.
- iii) Find the area of the pond?

iv) Find the area of the portion where grass is grown.

v) It is required to add a rectangular plot that is of area equal to the area of the portion where the grass is grown, with AD as a side outside the land. Draw a sketch of this rectangle with its measurements in the above figure.

3. (A) Mr. Perera's annual assessed value of the house is Rs. 950 000 and his annual income is Rs. 750 000. The local authority charges 8% of the assessed annual value of the house as rates.

(i) How much has to be paid as rates for year?

(ii) How much has to be paid as rates for a quarter?

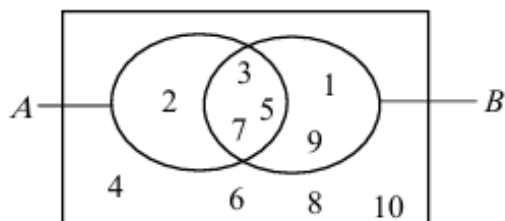
(B) The first Rs. 500 000 of annual income is exempt from income tax and the next Rs. 500 000 is subject to 4% income tax

(i) How much Mr. Perera's annual income tax?

(ii) Find his annual income after paying the rates for a year and income tax?

4. (a) $\varepsilon = \{\text{Integers from 1 to 10}\}$

(i) Write the set in description form within curly bracket.



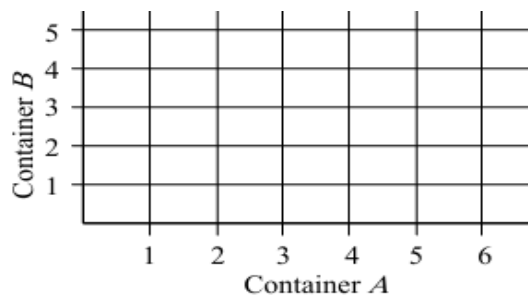
(ii) Write the set $A \cap B$ in description form.

(iii) Find $n(A' \cap B)$.

(b) There are six identical cards numbered 1, 2, 3, 4, 5, 6 in container A and five identical cards numbered 1, 2, 3, 4, 5 in container B. A child takes one card from each container and make a number. Here the number which is taken from A is put in the tens place and the number from the other is put to the unit place.

(i) Represent the sample space in a grid.

(ii) Find the probability of a number being a multiple of 11.





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Mathematics 2**

Answer five questions only.

1. An incomplete table of values prepared to draw the graph of the function $y = -x^2 + 7$ is given below.

x	-3	-2	-1	0	1	2	3
y	-2	3		7		3	-2

- a) i. Fill the blanks in the table.
 ii. By taking a suitable scale draw the graph of the above function on a graph paper.
- b) By using the graph,
 i. What is the maximum or minimum value?
 ii. Find the values of x when $y = 1.5$.
 iii. Write down the equation of the function when the above graph is shifted down 2 units along the y axis.

2. (a) Solve $8 = \left(\frac{1}{4}\right)^{2-x}$

- (b) Find the value without using the logarithmic table.

$$\frac{3}{2} \lg 25 + \frac{4}{5} \lg 32 - \lg 2$$

- (c) Find the value of P to the nearest first decimal place using logarithmic tables.

$$P = \frac{35.2 \times \sqrt[3]{0.538}}{1.25^2}$$

3. (a) It is spent Rs. 175 to buy two apples and five guavas. 10 guavas can be bought to the amount which is needed to buy 3 apples.
- (i) By taking the price of an apple as Rs. x and the price of a guava as Rs. y , build up a pair of simultaneous equation.
- (ii) Find separately the price of an apple and the price of a guava by solving the above pair of equation.

(b) $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$

Using the above result write the expansion of $(x - 2)^3$.

4. A hemispherical container with radius $2r$ is completely filled with water. Then this amount of water is poured into a cane with radius r and height $2r$ and after this cane was filled, the remaining water was poured into a cuboid with the base of square shaped. If the side of the square is a show that the water goes up to,

$$h = \frac{10\pi r^3}{a^2}$$

If $\pi = 3.14$, $r = 10.5\text{cm}$ and $a = 35\text{cm}$ by using logarithms find the value of h .

5. In the triangle ABC , $\hat{A}BC = 45^\circ$, $BC = 8\text{cm}$ and $\hat{B}AC = 60^\circ$
- Construct the triangle ABC .
 - Construct a perpendicular from A to BC . This perpendicular and the line BC meet at P . Mark the point P .
 - Construct the locus of the points which move equidistant to A and C . Mark the point it meets the line AC as R .
 - The above perpendicular in ii and the locus in iii meet at the point Q , Mark Q .
 - Show that, $\hat{Q}R + \hat{P}C = 180^\circ$.

6. In the trapezium $ABCD$ the diagonal AC bisected $\hat{B}AD$.
The bisector of $\hat{A}BC$ meet the diagonal AC at E .

- Prove that $AB = BC$.
- Prove that $\hat{A}EB = 90^\circ$.
- If $AC = AD$, then prove that $\hat{B}CD = 90 + \frac{1}{2}\hat{B}AC$

